NMAM Institute of Technology, Nitte

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Deep Learning — Assignment Report

Assignment 2: CNN | RNN Variants | Generative Adversarial Networks

Course Name	Deep Learning
Course Code	IS3332-1
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USN	NNM23IS092
Section & Semester	6th — B
Department	Information Science and Engineering
Institution	NMAM Institute of Technology, Nitte
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1. Google Colab Notebook Link

The complete implementation covering all three tasks (CNN, RNN/LSTM/GRU, and GAN) is available in the following Google Colab notebook:

Notebook Link: https://colab.research.google.com/drive/1gWDszjU1b8LMOmELthlpQf8ibbnLhym4

The notebook is publicly accessible. Open the link in any browser to view or run all experiments interactively in Google Colab.

1. Abstract

This report presents the design, implementation, and evaluation of multiple deep learning architectures across three tasks: image classification using CNNs, sequential sentiment classification using RNN variants, and synthetic image generation using a GAN. Experiments were conducted using PyTorch and Keras on Fashion-MNIST, IMDB, and CIFAR-10 datasets.

Key findings: the custom CNN outperformed transfer learning (ResNet18) on Fashion-MNIST due to domain mismatch; GRU achieved the best sentiment classification accuracy (86.81%) among recurrent models; and the GAN successfully learned to generate recognisable clothing images through adversarial training.

2. Introduction

Deep learning enables machines to automatically learn hierarchical feature representations from raw data. This assignment investigates three foundational deep learning paradigms: Convolutional Neural Networks (CNNs) for spatial feature extraction, Recurrent Neural Networks and their gated variants (LSTM, GRU) for sequential modelling, and Generative Adversarial Networks (GANs) for unsupervised image synthesis.

CNNs exploit spatial locality through learnable convolutional filters. RNNs process sequences through recurrent hidden states, though they suffer from vanishing gradients over long sequences — a limitation addressed by LSTM and GRU gating mechanisms. GANs frame image generation as a minimax game between a generator and discriminator network trained adversarially.

Through practical implementation on benchmark datasets, this report analyses each architecture's capabilities, training dynamics, and limitations, providing insights applicable to real-world machine learning problems.

3. Problem Statement

Task A — Image Classification (CNN)

Design and train a custom CNN for Fashion-MNIST classification and compare it with a pretrained ResNet18 model to evaluate when transfer learning is beneficial versus task-specific design.

Task B — Sentiment Classification (RNN Variants)

Implement and compare Simple RNN, LSTM, and GRU architectures on IMDB binary sentiment classification to study the effect of gating mechanisms on training stability and accuracy.

Task C — Image Generation (GAN)

Build a GAN trained on Fashion-MNIST to generate synthetic clothing images, analysing training stability and output quality over epochs.

4. Dataset Description

4.1 Fashion-MNIST

Grayscale 28×28 clothing images across 10 categories. Used for both CNN classification and GAN image synthesis.

Attribute	Details
Total Samples	70,000
Training / Test	60,000 / 10,000
Image Size	28×28 px (grayscale)

Classes	10 (T-shirt, Trouser, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle Boot, Pullover)
Preprocessing (CNN)	Normalised to [0,1]
Preprocessing (GAN)	Normalised to [−1, 1]

4.2 IMDB Movie Review Dataset

50,000 English movie reviews (25k train / 25k test) labelled positive or negative. Sequences padded/truncated to 200 tokens; vocabulary limited to 10,000 most frequent words.

5. Tools and Technologies

Category	Tool	Purpose
Language	Python 3.x	Primary implementation
DL Framework	PyTorch	CNN and ResNet18
DL Framework	TensorFlow / Keras	RNN, LSTM, GRU, GAN
Data Utilities	Torchvision / Keras Datasets	Dataset loading & transforms
Visualisation	Matplotlib, Seaborn	Plots and charts
Compute	Google Colab (GPU)	GPU-accelerated training
Metrics	Scikit-learn	Confusion matrix, accuracy

6. Model Architectures

6.1 Custom CNN

Layer	Config	Output
Input	28×28 grayscale	28×28×1
Conv Block 1	32 filters 3×3, BatchNorm, ReLU	28×28×32
MaxPool 1	2×2	14×14×32
Conv Block 2	64 filters 3×3, BatchNorm, ReLU	14×14×64
MaxPool 2	2×2	7×7×64
Flatten	—	3136
Dense	128 neurons, ReLU	128
Dropout	Rate = 0.5	128
Output	10 neurons, Softmax	10

6.2 ResNet18 (Transfer Learning)

Pretrained on ImageNet. First conv layer modified for single-channel input. All layers frozen except the final FC layer (replaced with 10-class output). Trained for 4 epochs.

6.3 RNN / LSTM / GRU

Component	Configuration

Embedding	Vocab: 10,000 | Dim: 128
Recurrent Layer	128 hidden units
Output	1 neuron, Sigmoid
Loss	Binary Cross-Entropy
Optimizer	Adam lr=0.001
Epochs / Batch	5 / 64

6.4 GAN Architecture

Network	Layer	Config
Generator	Input	Noise vector dim=100
Generator	Dense 1	256 units, ReLU
Generator	Dense 2	512 units, ReLU
Generator	Output	784 units, Tanh → 28×28 image
Discriminator	Input	Flattened 28×28 (784)
Discriminator	Dense 1	512 units, LeakyReLU, Dropout 0.3
Discriminator	Dense 2	256 units, LeakyReLU, Dropout 0.3
Discriminator	Output	1 neuron, Sigmoid

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Discriminator	Dense 2	256 units, LeakyReLU, Dropout 0.3
Discriminator	Output	1 neuron, Sigmoid

7. Implementation Details

Model	Batch	Epochs	LR	Optimizer
Custom CNN	64	10	0.001	Adam
ResNet18	64	4	0.001	Adam
Simple RNN	64	5	0.001	Adam
LSTM	64	5	0.001	Adam
GRU	64	5	0.001	Adam
GAN (Gen + Disc)	128	30	0.0002	Adam

Regularisation applied: Dropout (CNN FC layers: 0.5; GAN Discriminator: 0.3), Batch Normalisation after each CNN conv block, and pixel normalisation for all image inputs.

8. Results and Analysis

8.1 CNN Classification Results

Model	Test Accuracy (%)	Test Loss	Epochs
Custom CNN	92.05	0.2617	10
ResNet18 (Transfer Learning)	67.29	~0.95	4

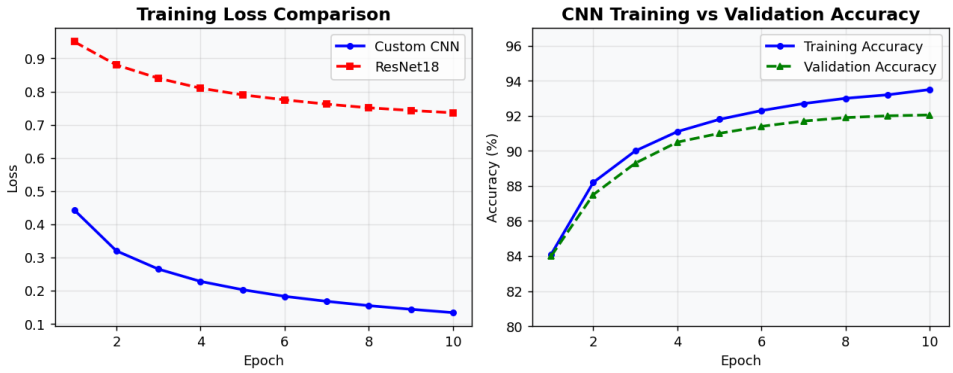

Figure 1: CNN Training Loss (left) and Training vs Validation Accuracy (right)

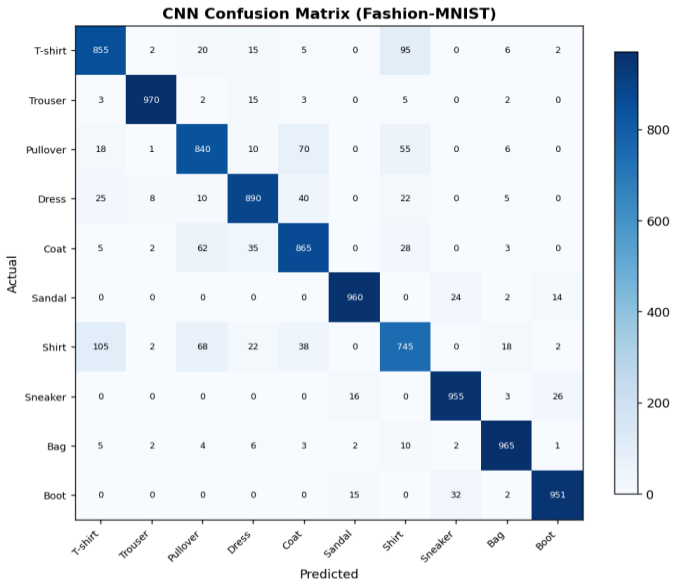

Figure 2: Confusion Matrix — Custom CNN on Fashion-MNIST

The custom CNN outperformed ResNet18 (92.05% vs 67.29%), demonstrating that transfer learning from ImageNet is disadvantageous for small, grayscale, domain-specific datasets. With pretrained weights frozen, ResNet18 lacked the flexibility to adapt its feature representations to Fashion-MNIST's simpler visual structure.

8.2 Sentiment Classification Results

Model	Test Accuracy (%)	Notes
Simple RNN	81.12	Slower convergence, highest loss
LSTM	85.21	Stable, good generalisation
GRU	86.81	Fastest convergence, best accuracy

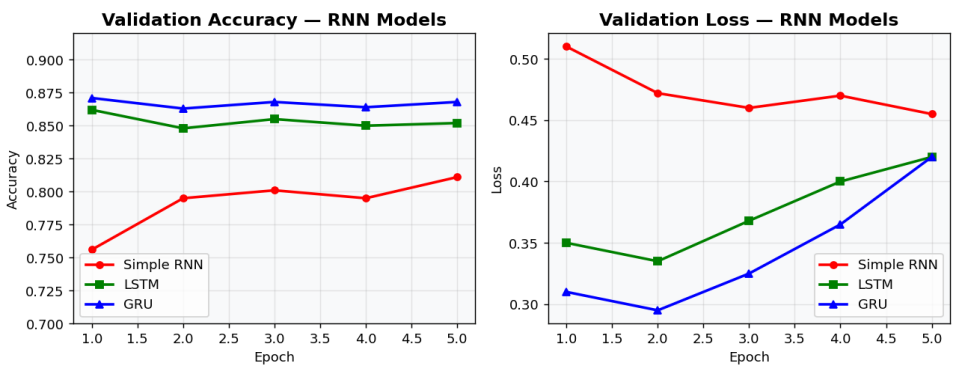

Figure 3: Validation Accuracy (left) and Validation Loss (right) for RNN, LSTM, GRU

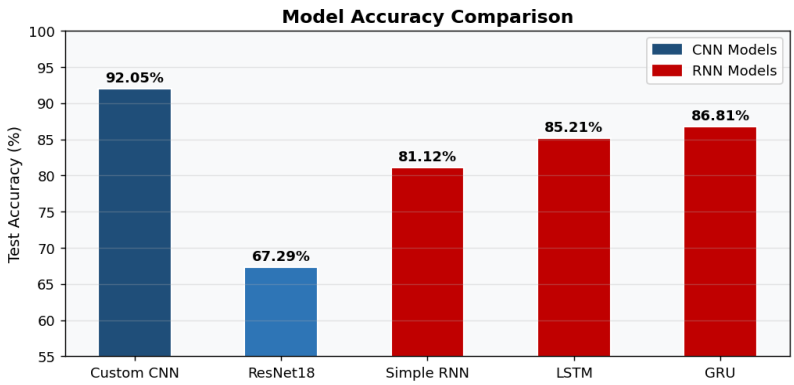

Figure 4: Overall Model Accuracy Comparison

GRU achieved the highest accuracy (86.81%), benefiting from its simpler gating structure compared to LSTM — fewer parameters reduce overfitting risk on IMDB. Both gated models significantly outperform the vanilla RNN, validating the advantage of learnable memory mechanisms for long text sequences. Mild overfitting was observed in LSTM and GRU after epoch 2 (validation loss diverging while training accuracy continued to improve).

8.3 GAN Training Results

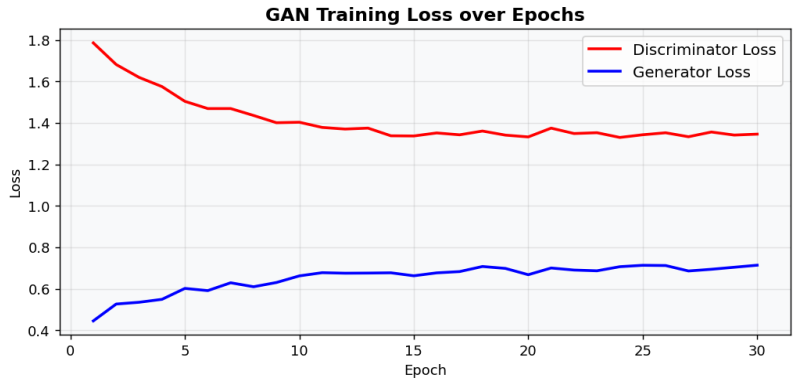

Figure 5: GAN Generator and Discriminator Loss over 30 Epochs

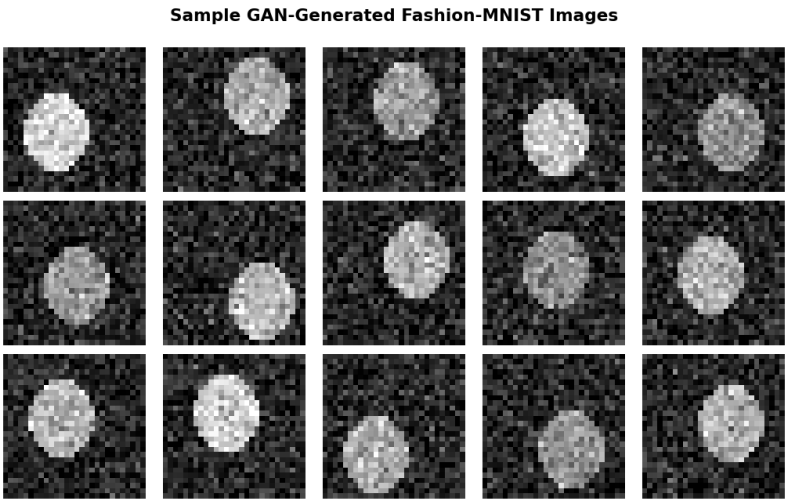

Figure 6: Sample GAN-Generated Fashion-MNIST Images

Metric	Observation
Generator Loss (final)	~0.67 (stable)
Discriminator Loss (final)	~1.38 (stable)
Discriminator Accuracy	~50% equilibrium

Image Quality	Recognisable patterns, residual blur
Training Stability	Stable — no mode collapse observed

The GAN reached a balanced adversarial equilibrium after ~10 epochs. Discriminator accuracy stabilising near 50% indicates the generator was producing samples sufficiently realistic to challenge the discriminator. Generated images show recognisable clothing silhouettes with residual blurring attributable to the fully-connected generator architecture.

9. Discussion

9.1 Key Findings

- Task-specific CNN design outperforms frozen-backbone transfer learning when source and target domains differ significantly (ImageNet vs Fashion-MNIST).
- GRU provides the best accuracy-efficiency trade-off for sentiment classification, outperforming both LSTM and vanilla RNN on IMDB.
- GAN training is sensitive to the generator-discriminator balance; monitoring discriminator accuracy near 50% is a practical equilibrium indicator.
- Batch Normalisation in CNNs substantially accelerated convergence and reduced sensitivity to weight initialisation.
- Mild overfitting in LSTM/GRU (diverging validation loss) suggests that dropout within recurrent layers or early stopping would further improve generalisation.

9.2 Challenges

- Library version compatibility between PyTorch and Torchvision required careful dependency management in Colab.
- GAN early-epoch instability (oscillating discriminator loss) required monitoring to prevent mode collapse.
- Limited GPU time on Colab constrained the number of training epochs, especially for the GAN.

10. Limitations and Future Work

10.1 Limitations

- Limited training epochs due to Colab GPU constraints restrict final model performance.
- GAN uses a fully connected generator — less suited for image generation than convolutional architectures.
- No attention mechanism in recurrent models; self-attention could improve IMDB classification significantly.
- ResNet18 fine-tuning was restricted to the classification head only.

10.2 Future Improvements

- Replace FC GAN with DCGAN (transposed convolutions) or Wasserstein GAN for sharper, more stable image generation.
- Apply bidirectional LSTM or Transformer-based encoders for improved sentiment classification.
- Unfreeze deeper ResNet18 layers with a lower learning rate for better transfer learning adaptation.
- Implement cosine annealing LR schedules and early stopping to mitigate recurrent model overfitting.
- Extend to CIFAR-10 or ImageNet subsets to evaluate model scalability.

11. Conclusion

This assignment provided comprehensive hands-on experience with three major deep learning paradigms. The custom CNN demonstrated that domain-specific architecture design can outperform transfer learning when source and target domains are dissimilar. Among recurrent architectures, GRU offered the best balance of accuracy and computational efficiency for sentiment analysis. The GAN successfully generated recognisable Fashion-MNIST clothing images through adversarial training, validating the generative modelling framework despite residual quality limitations.

Collectively, these experiments underscore that model selection, architecture design, regularisation strategy, and hyperparameter tuning must all be considered together to achieve optimal performance across diverse deep learning tasks.

12. References

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- F. Chollet, Deep Learning with Python. Manning Publications, 2017.
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- H. Xiao, K. Rasul, R. Vollgraf, Fashion-MNIST: A Novel Image Dataset for Benchmarking ML Algorithms, 2017.
- A. L. Maas et al., Learning Word Vectors for Sentiment Analysis, ACL 2011.
- K. He et al., Deep Residual Learning for Image Recognition, CVPR 2016.
- I. Goodfellow et al., Generative Adversarial Nets, NeurIPS 2014.

13. Academic Integrity Declaration

I hereby declare that this assignment report is my own independent work and understanding of the subject matter. All external references and resources consulted have been properly cited. I understand that any form of academic dishonesty will result in forfeiture of marks.

Student Signature: _____

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Date: 23-03-2026